

Antibiotic-Associated Complications and Resistance

Gastrointestinal disturbances and oral mucous membrane lesions are the major problems encountered with alteration of the normal microbial flora. There are numerous other potential untoward reactions to antibiotics—such as renal, hematologic, hepatic, and nervous system toxicity—that may represent acceptable risks if the indications for antibiotic use are compelling. It is the physician's responsibility to be aware of both the common and rare manifestations of antibiotic toxicity and to remain alert for novel adverse effects in individual patients.

Antibiotic Resistance Due to "R" Factors

Transferable resistance to multiple antibiotics has become a widespread clinical challenge. This resistance is largely mediated by extrachromosomal genetic elements known as R plasmids. Prolonged antibiotic therapy can select for R-factor-carrying strains within the indigenous flora, particularly in the gastrointestinal tract. These resistant bacteria may then transfer their resistance genes to potential pathogens during conjugation. For example, an R plasmid in *E. coli* that confers resistance to chloramphenicol, tetracycline, and kanamycin can be transferred to *Klebsiella* or *Salmonella* species during bacterial mating.

Topical Antibacterial Agents

Topical antibacterial agents are commonly used to prevent or suppress bacterial growth in open lesions and surgical wounds. However, in sutured dermatologic wounds, topical antibiotics have limited efficacy in reducing infection rates. Topical neomycin and bacitracin frequently cause contact dermatitis, particularly when applied to venous insufficiency ulcers of the lower extremities.

Among the most effective topical antibacterial agents are: - **Acetic acid (1% to 5%)**: Useful for *Pseudomonas* infections involving the nails and toe web spaces. - **Bacitracin (500 units per milliliter or gram)**: Effective against superficial infections caused by *Staphylococcus aureus* and streptococci. - **Neomycin (0.5% ointment)** and **gentamicin (0.17% cream)**: May be beneficial in selected cases requiring local suppression of mixed Gram-negative bacteria. - **Mupirocin (2% ointment)**: Active against various streptococci and *S. aureus*, including some methicillin-resistant strains.

Several broad-spectrum antiseptics are also available for topical use: - **Povidone-iodine (Betadine)**: Effective against most Gram-positive and Gram-negative bacteria, though it lacks persistent residual activity on the skin. - **Chlorhexidine gluconate (4% solution)**: Offers broad antibacterial coverage with prolonged action. An alcohol-based formulation is particularly effective, is minimally absorbed systemically, and is generally non-irritating. These antiseptics can be used prophylactically or to treat superficially infected dermatoses and minor wounds.

Key References

The full reference list for all chapters is available at www.digm7.com.

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